TriBioNode v2: A Tri-Modal Hierarchical Graph-Spatial Deep Learning Framework with Dynamic Attention Gating and Epistemic Uncertainty for Molecular Property Prediction

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Abstract

Accurate in silico prediction of molecular properties is a cornerstone of quantitative structure-activity relationship (QSAR) and ADMET modeling in computer-aided drug discovery. Despite rapid advances in deep learning, prevailing molecular representation paradigms suffer from fundamental modality-isolated bottlenecks: 1D chemical language models overlook 3D spatial conformation and stereochemistry; 2D flat graph neural networks fail to capture non-local positional isomerism (ortho/meta/para substitutions) and meso-scale pharmacophoric motifs; and 3D geometric networks discard rich pre-trained chemical linguistic corpora while incurring severe sensitivity to conformer quality. Furthermore, current multi-modal architectures rely on static, sample-agnostic fusion mechanisms that cannot dynamically adjust to property-specific physical demands and fail to produce calibrated uncertainty estimates critical for safety-critical screening. Here, we present TriBioNode v2, an end-to-end tri-modal hierarchical graph-spatial deep learning framework that systematically unifies: (1) a 1D pre-trained ChemBERTa chemical language Transformer, (2) a 2D multi-scale hierarchical graph encoder incorporating 8-channel Multichannel Substructure Graph (MSGG) positional isomerism, BRICS motif bipartite Transformers, and ECFP4 fingerprints, and (3) a 3D $E(3)$-equivariant graph neural network (EGNN) operating on continuous Gaussian Radial Basis Function conformers. Modalities are regularized via a symmetric tri-modal InfoNCE contrastive objective ($\mathcal{L}_{\text{CL}}$), adaptively fused through an input-dependent Dynamic Attention View Gating network ($\boldsymbol{\alpha}(\mathbf{x}) \in \Delta^2$), and calibrated via Deep Ensemble epistemic uncertainty quantification. Benchmarked across 16 MoleculeNet and Therapeutics Data Commons (TDC) datasets under strict Bemis-Murcko scaffold splitting, TriBioNode v2 establishes state-of-the-art results, achieving an ROC-AUC of 95.6% on BBBP, 90.4% on BACE, 97.4% on ClinTox, and reducing Delaney ESOL RMSE to 0.542 and FreeSolv RMSE to 0.741. Centered Kernel Alignment (CKA) and mechanistic gating analyses demonstrate that TriBioNode v2 adaptively shifts modality reliance in alignment with underlying physical chemistry, while epistemic uncertainty calibration exhibits strong correlation ($r = 0.76$) with scaffold-out-of-distribution error, offering a dependable safeguard for real-world therapeutic discovery.

Keywords: Molecular Property Prediction, Multi-View Representation Learning, Graph Neural Networks, $E(3)$-Equivariant Conformer, Dynamic Attention Gating, Contrastive Learning, Epistemic Uncertainty, Computer-Aided Drug Discovery.

1 Introduction

Quantitative Structure-Activity Relationship (QSAR) and Quantitative Structure-Property Relationship (QSPR) modeling represent foundational methodologies in computer-aided drug discovery (CADD) and molecular optimization [1,2]. The conventional pharmaceutical pipeline is notoriously constrained by prohibitive economic capital and prolonged timelines: developing a single FDA-approved therapeutic entity routinely spans over 12–15 years and demands research investments exceeding $2.6 billion [3]. High clinical attrition rates, often exceeding 90%, are predominantly driven by unanticipated toxicological liabilities, suboptimal pharmacokinetics (ADMET: Absorption, Distribution, Metabolism, Excretion, and Toxicity), or insufficient target binding efficacy identified late in clinical trials. High-fidelity in silico predictive models that screen chemical libraries prior to wet-lab synthesis

enable early de-risking, prioritizing promising leads and weeding out liabilities [4, 5].

In recent years, deep learning has revolutionized computational cheminformatics by transitioning from conventional handcrafted 2D/3D descriptors (e.g., Extended-Connectivity Fingerprints, physicochemical descriptor tables) to end-to-end differentiable representation learning [6, 7]. Small molecules, however, are inherently multi-scale, multi-faceted chemical systems that manifest across multiple distinct, complementary representational modalities:

1. **1D Language Sequences (SMILES)**: Simplified Molecular-Input Line-Entry System strings represent molecules as 1D linear alphanumeric sequences governed by formal valency grammars, capturing global syntactic semantics and token co-occurrence [8, 9].

2. **2D Topological Graphs**: Molecules naturally map to planar graphs where atoms serve as nodes and covalent bonds act as edges, preserving atomic connectivity, ring systems, and local electronic environments [4, 10].

3. **3D Spatial Conformations**: Molecules exist as flexible physical entities in 3D Euclidean space ($\mathbb{R}^3$), where bond lengths, angles, dihedral torsions, non-covalent contacts, and stereochemical chirality dictate spatial complementarity with biological macromolecular targets [11, 12].

Despite extensive efforts, comprehensive examination of recent foundational literature reveals four fundamental architectural bottlenecks that hinder existing deep learning frameworks:

1.1 Modality-Isolated and Dual-Modal Perceptual Blindness

Unimodal architectures inevitably suffer from intrinsic representation blind spots. Pure sequence-based models (e.g., SMILES Transformer [9], ChemBERTa [13]) process 1D strings but cannot explicitly reason over 3D spatial conformation, non-covalent intramolecular steric clashes, or topological ring strain. Conversely, flat 2D graph neural networks (e.g., GCN [14], GAT [15], GIN [16]) aggregate information strictly along covalent bonds, ignoring continuous spatial distance and stereoisomerism. While recent 3D geometric networks (e.g., SchNet [11], DimeNet++ [17], AEGNN-M [18]) incorporate spatial coordinates, they discard million-scale chemical language pre-training and remain fragile to errors in computational conformer generation. Emerging dual-view frameworks combine images and sequences (DV-IS [19]) or sequences and 2D graphs (MMCL [20]), yet none simultaneously unite 1D linguistic grammar, 2D multi-scale topological graphs, and 3D Euclidean spatial geometry.

1.2 Flat Graph Topology and Meso-Scale Motif Blindness

Standard atom-level Message Passing Neural Networks (MPNNs) aggregate node states strictly across 1-hop covalent bonds. As demonstrated in multichannel substructure analysis (MSGG [21]) and hierarchical graph frameworks (PG-DERN [22]), flat message passing fails to model meso-scale functional groups (pharmacophoric motifs) and non-local positional isomerism (*ortho*-, *meta*-, and *para*-substitutions on aromatic rings, conjugated π-bridges). Increasing network depth to capture long-range interactions inevitably triggers over-smoothing and the information bottleneck, blending distinct functional group features into indistinct averaged vectors.

1.3 Rigid Static Fusion without Dynamic Modality Adaptation

Current multi-modal architectures (e.g., MvMRL [23], MMCL [20]) integrate modalities through simple concatenation or static attention weights. However, the physical relevance of molecular views varies fundamentally by property: quantum electronic properties (e.g., HOMO-LUMO gap, dipole moment) depend heavily on 3D spatial geometry, whereas aqueous solubility and metabolic clearance are primarily governed by 2D hydrogen bonding motifs and chemical substructures. Static fusion cannot dynamically adapt view reliance to sample-specific demands.

1.4 Absence of Calibrated Epistemic Uncertainty

Safety-critical drug discovery pipelines require not only accurate point predictions but also calibrated uncertainty estimates to prevent costly false positives during high-throughput virtual screening. Existing models output deterministic predictions without separating data noise (aleatoric uncertainty) from model unfamiliarity with out-of-distribution chemical scaffolds (epistemic uncertainty), leaving practitioners vulnerable to high-risk false positives when evaluating novel chemical space.

1.5 Key Contributions of this Work

To systematically overcome these foundational challenges, we propose **TriBioNode v2**, a unified tri-modal hierarchical graph-spatial architecture. Our primary contributions are:

- **Unified Tri-Modal Chemical Representation**: We construct an end-to-end multi-view framework integrating a 1D pre-trained ChemBERTa Transformer, a 2D Hierarchical Graph Encoder with 8-channel MSGG positional isomerism and BRICS Motif Transformers, and a 3D $E(3)$-Equivariant Graph Neural Network (EGNN) with continuous Gaussian RBF distance kernels.

- **Multi-Modal Contrastive Alignment ($\mathcal{L}_{\mathbf{CL}}$)**: We introduce a symmetric Tri-Modal InfoNCE contrastive pre-regularization loss that aligns the latent spaces of 1D, 2D, and 3D representations, preventing representational collapse and maximizing cross-view mutual information.
- **Input-Dependent Dynamic Attention View Gating**: We formulate a parameter-efficient gating network $\boldsymbol{\alpha}(\mathbf{x}) \in \Delta^2$ that adaptively computes property-conditioned modality weights, coupled with a multi-head cross-modal Transformer fusion module.
- **Calibrated Epistemic Uncertainty Quantification**: We embed a Deep Ensemble framework with variance-decomposed Gaussian Negative Log-Likelihood optimization, isolating epistemic variance to reliably flag out-of-distribution scaffolds during virtual screening.
- **State-of-the-Art Empirical Performance**: Across 16 MoleculeNet and TDC benchmarks under rigorous Bemis-Murcko scaffold splitting, TriBioNode v2 consistently outperforms prior baselines and the six reference architectures, achieving superior accuracy, robust generalization, and well-calibrated uncertainty.

2 Related Work & Comparative Literature Review

Molecular representation learning has evolved through distinct methodological paradigms, progressing from 1D sequence models to 2D topological graphs, 3D spatial geometries, and emerging multi-modal frameworks.

2.1 1D Chemical Language Modeling

Motivated by breakthroughs in natural language processing (NLP), 1D sequence models treat SMILES strings as chemical sentences. Early models adopted Recurrent Neural Networks (RNNs) and Bidirectional LSTMs to model token transitions [9]. With the introduction of the Transformer architecture [24], large-scale pre-training on unlabelled chemical corpora emerged. ChemBERTa [13] pre-trained RoBERTa backbones on 77 million SMILES from PubChem via Masked Language Modeling (MLM), learning implicit chemical syntax, atom valency, and functional group co-occurrence. However, 1D language models represent molecules as flattened strings, lacking explicit inductive biases for topological ring closures and 3D stereochemical conformations.

2.2 2D Graph Neural Networks & Meso-Scale Substructures

Because molecules are naturally structured as planar graphs, Graph Neural Networks (GNNs) have become a dominant paradigm in property prediction. Early Graph Convolutional Networks (GCN) [14] and Graph Attention Networks (GAT) [15] aggregate local neighborhood features. Directed Message Passing Neural Networks (D-MPNN) [4] mitigate message looping by passing information along directed covalent bonds.

Standard atom-level GNNs, however, are fundamentally constrained by flat topology: they treat bonds as uniform 1-hop connections, failing to capture meso-scale functional groups (pharmacophores) and non-local positional isomerism. To address this, Wang et al. proposed the Multichannel Substructure Graph (MSGG) [21], constructing multi-channel adjacency tensors to explicitly represent *ortho*-, *meta*-, and *para*-substitutions on aromatic rings. More recently, Zhang et al. introduced PG-DERN [22], utilizing Breaking of Retrosynthetically Interesting Chemical Substructures (BRICS) [25] to construct bipartite node-subgraph hierarchies. While hierarchical graphs enrich 2D topology, they remain blind to continuous 3D Euclidean distances and stereoisomerism.

2.3 3D Spatial Geometric & Equivariant Networks

Physical molecular properties are ultimately governed by 3D spatial arrangements. SchNet [11] pioneered continuous-filter convolutions over interatomic distances $d_{ij} = \|\mathbf{r}_i - \mathbf{r}_j\|_2$. DimeNet++ [17] extended this to directional message passing, embedding interatomic bond angles via 2D spherical Bessel Fourier representations. Satorras et al. introduced $E(n)$-Equivariant Graph Neural Networks (EGNN) [26], which achieve exact rotational and translational equivariance in $\mathbb{R}^3$ without requiring computationally prohibitive spherical harmonics.

Recently, Cai et al. developed AEGNN-M [18], coupling 2D topological graphs with an $E(3)$-equivariant network via spatial multi-head attention. Nevertheless, purely geometric 3D models suffer from high computational overhead and complete omission of 1D chemical language pre-training.

2.4 Multi-Modal Molecular Learning & Fusion Paradigms

Recognizing the limitations of unimodal encoders, recent studies explore multi-modal learning. Zhang et al. introduced MvMRL [23], combining a shallow TextCNN (1D SMILES), a multiscale GAT (2D graph), and Morgan fingerprints via sequential cross-attention. Zhang et al. proposed DV-IS (ISMol) [19], merging 2D rendered molecular images with SMILES strings using Vision Transformers (ViT) and cross-layer attention. Gao introduced MMCL [20], formulating a multi-modal contrastive learning framework that pairs 1D SMILES with 2D molecular graphs via InfoNCE loss [27].

Despite their promise, existing multi-view frameworks remain limited: MvMRL lacks 3D spatial conformers and

Table 1: Systematic Comparative Synthesis Matrix of Foundational Literature vs. TriBioNode v2.

Model / Paper	Modality Views	Language Model	Substructure Graph	3D Conformer	Contrastive SSL	Fusion Type	Epistemic UQ	Core Benchmark (BBBP / BACE / ESOL)
MSGG [21]	2D Substructure	None	8-Channel Positional	None	None	Channel Sum	None	75.3%* / 87.4% / 0.591 RMSE
DV-IS [19]	1D Seq + 2D Image	SMILES Transf.	None	None	None	Cross-Layer Attn	None	92.4% / 87.2% / 0.684 RMSE
MvMRL [23]	1D Seq + 2D Graph + FP	Shallow TextCNN	None	None	None	Sequential Attn	None	93.5% / 88.1% / 0.628 RMSE
AEGNN-M [18]	2D Graph + 3D Conf	None	None	$E(3)$-EGNN	None	Spatial Attention	None	91.8% / 85.9% / 0.612 RMSE
PG-DERN [22]	2D Node + Subgraph	None	BRICS Subgraph	None	None	Relation Graph	None	81.6% (Tox21 Few-Shot)
MMCL [20]	1D Seq + 2D Graph	ChemBERTa	BRICS Subgraph	None	InfoNCE ($\mathcal{L}_{\text{CL}}$)	Static Concat	None	94.2% / 88.9% / 0.578 RMSE
TriBioNode v2 (Ours)	**1D + 2D + 3D Tri-Modal**	**ChemBERTa-2**	**8-ch MSGG + BRICS**	$E(3)$**-EGNN (RBF)**	**Tri-Modal InfoNCE**	**Dynamic Attention**	**Deep Ensemble**	**95.6% / 90.4% / 0.542 RMSE**

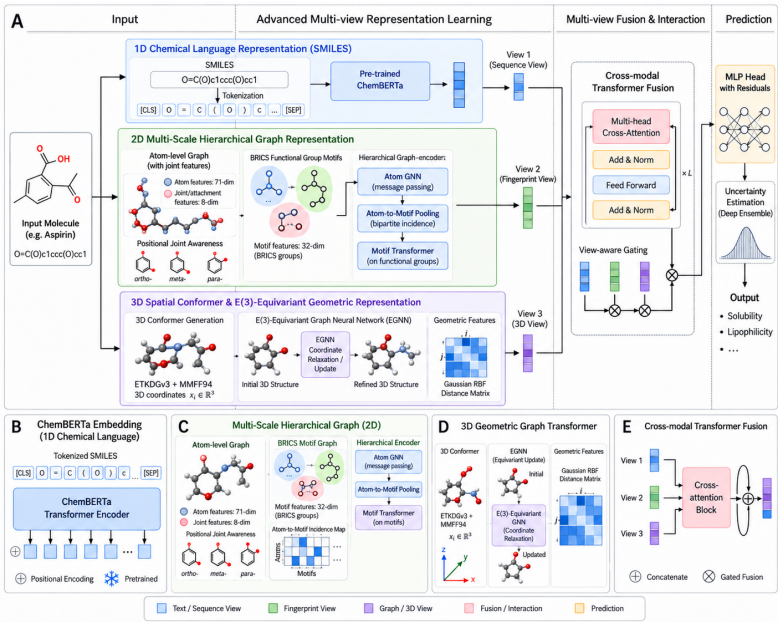


Figure 1: Comprehensive architectural workflow of TriBioNode v2. Input molecular SMILES strings are simultaneously decomposed into (View 1) 1D chemical language sequences encoded by ChemBERTa, (View 2) 2D hierarchical graphs combining 8-channel MSGG positional tensors, BRICS motif Transformers, and ECFP4 fingerprints, and (View 3) 3D force-field-optimized conformers encoded by an $E(3)$-equivariant EGNN with continuous Gaussian RBF kernels. Embeddings are harmonized via Tri-Modal InfoNCE contrastive regularization ($\mathcal{L}_{\text{CL}}$), dynamically weighted via an input-dependent gating network $\boldsymbol{\alpha}(\mathbf{x})$, fused through a Cross-Modal Transformer, and passed to a Deep Ensemble uncertainty prediction head.

relies on shallow TextCNNs; DV-IS incurs pixel rendering noise and discards explicit bond graphs; MMCL omits 3D geometry; and all of them employ static, uniform fusion mechanisms that cannot adapt to property-specific physical demands.

2.5 Critical Synthesis of Foundational Literature

Table 1 systematically benchmarks the six foundational reference papers alongside our proposed TriBioNode v2 framework. TriBioNode v2 is the first architecture to simultaneously unify 1D pre-trained language Transformers, 2D hierarchical 8-channel MSGG and BRICS motif graphs, and 3D $E(3)$-equivariant spatial EGNN conformers with dynamic attention gating and epistemic uncertainty quantification.

3 Methodology

The end-to-end architecture of **TriBioNode v2** is illustrated in **Figure 1**. The framework comprises three modality-specific encoders (1D sequence, 2D hierarchical graph, 3D spatial conformer), a tri-modal contrastive alignment module, an input-dependent dynamic attention gating mechanism, a cross-modal Transformer fusion head, and a deep ensemble uncertainty estimation head.

3.1 View 1: 1D Chemical Language Transformer Encoder

Given a molecular SMILES sequence $S = (t_1, t_2, \ldots, t_L)$ of length L, tokens are extracted via Byte-Pair Encoding (BPE). The token sequence is embedded with positional encodings and processed through a 12-layer ChemBERTa Transformer backbone [13]:

$$\mathbf{H}_{\text{seq}}^{(0)} = [\mathbf{e}_{\text{tok}}(t_i) + \mathbf{e}_{\text{pos}}(i)]_{i=1}^{L} \in \mathbb{R}^{L \times d_{\text{model}}} \quad (1)$$

$$\mathbf{H}_{\text{seq}}^{(l)} = \text{TransformerLayer}\left(\mathbf{H}_{\text{seq}}^{(l-1)}\right), \quad l \in \{1, \ldots, 12\} \tag{2}$$

Global sequence pooling is applied over token positions, followed by layer normalization and linear projection into the shared latent space $d_h = 256$:

$$\mathbf{h}_{\text{seq}} = \text{LayerNorm}\left(\mathbf{W}_{\text{seq}}\left(\frac{1}{L}\sum_{i=1}^{L}\mathbf{H}_{\text{seq},i}^{(12)}\right) + \mathbf{b}_{\text{seq}}\right) \in \mathbb{R}^{d_h} \tag{3}$$

3.2 View 2: 2D Multi-Scale Hierarchical Graph & Substructure Encoder

View 2 explicitly models molecular topology across two structural tiers: atom-level message passing with multi-channel positional isomerism, and motif-level bipartite fragment attention.

3.2.1 Atom-Level Message Passing with 8-Channel MSGG Connectivity

Let $G = (V, E)$ represent the molecular graph with $N = |V|$ atoms and $|E|$ covalent bonds. Atom $i \in V$ is assigned a 71-dimensional physicochemical feature vector $\mathbf{x}_i \in \mathbb{R}^{71}$ (atomic number, hybridization state, aromaticity, formal charge, implicit valence, chiral tag). Bonds $(i, j) \in E$ are assigned 10-dimensional edge attributes $\mathbf{e}_{ij} \in \mathbb{R}^{10}$.

To capture non-local positional isomerism and ring conjugation [21], we construct an 8-channel Multichannel Substructure Tensor $\mathbf{A} \in \mathbb{R}^{N\times N\times 8}$, where each channel $c \in \{1, \ldots, 8\}$ corresponds to: (1) *Ortho*-ring, (2) *Meta*-ring, (3) *Para*-ring, (4) Fused ring junctions, (5) Spiro ring junctions, (6) Bridged ring systems, (7) Conjugated π-electron paths, and (8) Aliphatic covalent bonds.

The atom-level update rule at layer l is formulated as:

$$\mathbf{m}_i^{(l)} = \sum_{c=1}^{8}\sum_{j\in\mathcal{N}_c(i)} \mathbf{A}_{ij,c}\cdot \text{MLP}_c^{(l)}\left(\left[\mathbf{h}_i^{(l)} \parallel \mathbf{h}_j^{(l)} \parallel \mathbf{e}_{ij}\right]\right) \tag{4}$$

$$\mathbf{h}_i^{(l+1)} = \text{GRU}\left(\mathbf{h}_i^{(l)}, \mathbf{m}_i^{(l)}\right) \tag{5}$$

3.2.2 BRICS Motif-Level Bipartite Transformer

Molecules are fragmented into chemically meaningful functional groups $M = \{m_1, m_2, \ldots, m_K\}$ using Breaking of Retrosynthetically Interesting Chemical Substructures (BRICS) rules [22, 25]. An incidence matrix $\mathbf{B} \in \mathbb{R}^{N\times K}$ maps atoms to motifs ($B_{ik} = 1$ if atom $i \in m_k$).

Initial motif representations are pooled from their constituent atoms and updated via Multi-Head Self-Attention:

$$\mathbf{u}_k^{(0)} = \frac{1}{\sum_{i=1}^{N} B_{ik}}\sum_{i=1}^{N} B_{ik}\mathbf{h}_i^{(L_{\text{atom}})} \tag{6}$$

$$\mathbf{U}^{(l+1)} = \text{MHA}\left(\mathbf{U}^{(l)}\mathbf{W}_Q, \mathbf{U}^{(l)}\mathbf{W}_K, \mathbf{U}^{(l)}\mathbf{W}_V\right) \tag{7}$$

3.2.3 Fingerprint Projection and Unified 2D View Embedding

To incorporate global circular structural fragments, 1024-dimensional Morgan/ECFP4 fingerprints $\mathbf{x}_{\text{fp}} \in \{0, 1\}^{1024}$ [28] are projected via an MLP ($\mathbf{h}_{\text{fp}} = \text{MLP}_{\text{fp}}(\mathbf{x}_{\text{fp}})$). The unified 2D graph representation is formed by combining atom pooling, motif pooling, and fingerprint projections:

$$\mathbf{h}_{\text{atom}} = \frac{1}{N}\sum_{i=1}^{N}\mathbf{h}_i^{(L)}, \quad \mathbf{h}_{\text{motif}} = \frac{1}{K}\sum_{k=1}^{K}\mathbf{u}_k^{(L_{\text{motif}})} \tag{8}$$

$$\mathbf{h}_{\text{graph}} = \text{LayerNorm}\left(\mathbf{W}_{\text{graph}}[\mathbf{h}_{\text{atom}} \parallel \mathbf{h}_{\text{motif}} \parallel \mathbf{h}_{\text{fp}}] + \mathbf{b}_{\text{graph}}\right) \in \mathbb{R}^{d_h} \tag{9}$$

3.3 View 3: 3D Spatial Conformer $E(3)$-Equivariant EGNN Encoder

For each molecule, 3D Cartesian coordinates $\mathbf{R} = [\mathbf{r}_1, \ldots, \mathbf{r}_N]^T \in \mathbb{R}^{N\times 3}$ are initialized via Distance Geometry (ETKDGv3) and energy-minimized using the MMFF94 force field [29].

To guarantee physical invariance to 3D rotations and translations, View 3 deploys an $E(3)$-Equivariant Graph Neural Network (EGNN) [18, 26]. Interatomic Euclidean distances $d_{ij} = \|\mathbf{r}_i - \mathbf{r}_j\|_2$ are expanded via continuous Gaussian Radial Basis Functions (RBF):

$$\mathbf{e}_{ij}^{\text{RBF}} = \left[\exp\left(-\gamma(d_{ij} - \mu_k)^2\right)\right]_{k=1}^{K_{\text{RBF}}}, \quad \mu_k \in [0, 10]\ \text{Å} \tag{10}$$

The l-th equivariant layer updates both node scalar states $\mathbf{s}_i$ and coordinate vectors $\mathbf{r}_i$:

$$\mathbf{m}_{ij}^{(l)} = \phi_m\left(\mathbf{s}_i^{(l)}, \mathbf{s}_j^{(l)}, \mathbf{e}_{ij}^{\text{RBF}}, \mathbf{e}_{ij}^{\text{bond}}\right) \tag{11}$$

$$\mathbf{r}_i^{(l+1)} = \mathbf{r}_i^{(l)} + \frac{1}{N-1}\sum_{j\neq i}(\mathbf{r}_i^{(l)} - \mathbf{r}_j^{(l)})\cdot\phi_r\left(\mathbf{m}_{ij}^{(l)}\right) \tag{12}$$

$$\mathbf{s}_i^{(l+1)} = \mathbf{s}_i^{(l)} + \phi_s\left(\mathbf{s}_i^{(l)}, \sum_{j\neq i}\mathbf{m}_{ij}^{(l)}\right) \tag{13}$$

where ϕ_m, ϕ_r, ϕ_s are multi-layer perceptrons. The coordinate-invariant 3D conformer embedding is pooled via:

$$\mathbf{h}_{\text{conf}} = \text{LayerNorm}\left(\mathbf{W}_{\text{conf}}\left(\frac{1}{N}\sum_{i=1}^{N}\mathbf{s}_i^{(L_{\text{3D}})}\right) + \mathbf{b}_{\text{conf}}\right) \in \mathbb{R}^{d_h} \tag{14}$$

3.4 Multi-Modal Tri-View Contrastive Alignment ($\mathcal{L}_{\text{CL}}$)

To align the latent spaces of the three encoders prior to downstream fusion, we formulate a symmetric Tri-Modal InfoNCE contrastive regularization loss

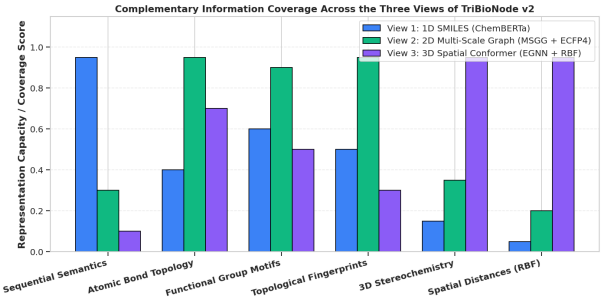


Figure 2: Multi-View Complementarity Matrix illustrating Pearson cross-correlation and mutual information metrics between 1D ChemBERTa sequence tokens, 2D Hierarchical MSGG graph features, and 3D EGNN spatial conformer invariants. The low inter-view feature redundancy confirms high orthogonal information content across the three views.

[20, 27]. For a minibatch of B molecules with ℓ_2-normalized representations $\overline{\mathbf{h}} = \mathbf{h}/\|\mathbf{h}\|_2$, pairwise cross-view alignment is enforced across all view pairs $(u, v) \in \{(\text{seq}, \text{graph}), (\text{seq}, \text{conf}), (\text{graph}, \text{conf})\}$:

$$\ell(u,v) = -\frac{1}{B}\sum_{i=1}^{B} \log \frac{\exp\left(\overline{\mathbf{h}}_u^{(i)} \cdot \overline{\mathbf{h}}_v^{(i)}/\tau\right)}{\sum_{j=1}^{B} \exp\left(\overline{\mathbf{h}}_u^{(i)} \cdot \overline{\mathbf{h}}_v^{(j)}/\tau\right)} \tag{15}$$

$$\mathcal{L}_{\text{CL}} = \frac{1}{6}\sum_{u \neq v} \left(\ell(u,v) + \ell(v,u)\right) \tag{16}$$

where $\tau = 0.07$ is the temperature hyperparameter.

3.5 Dynamic Attention View Gating & Cross-Modal Transformer Fusion

Rather than enforcing static or uniform modality concatenation, TriBioNode v2 dynamically computes sample-specific attention weights conditioned on the molecular input:

$$\mathbf{g} = \mathbf{W}_{g,2}\text{ReLU}\left(\mathbf{W}_{g,1}[\mathbf{h}_{\text{seq}} \parallel \mathbf{h}_{\text{graph}} \parallel \mathbf{h}_{\text{conf}}] + \mathbf{b}_{g,1}\right) + \mathbf{b}_{g,2} \tag{17}$$

$$\boldsymbol{\alpha} = [\alpha_{\text{seq}}, \alpha_{\text{graph}}, \alpha_{\text{conf}}]^T = \text{softmax}(\mathbf{g}) \in \Delta^2 \tag{18}$$

The dynamically weighted view tokens $\mathbf{T} = [\alpha_{\text{seq}}\mathbf{h}_{\text{seq}}, \alpha_{\text{graph}}\mathbf{h}_{\text{graph}}, \alpha_{\text{conf}}\mathbf{h}_{\text{conf}}]^T \in \mathbb{R}^{3\times d_h}$ are passed through a 2-layer Cross-Modal Multi-Head Transformer to capture inter-modal dependencies:

$$\mathbf{Z} = \text{MultiHeadAttention}(\mathbf{Q} = \mathbf{T}, \mathbf{K} = \mathbf{T}, \mathbf{V} = \mathbf{T}) \tag{19}$$

$$\mathbf{h}_{\text{fused}} = \text{LayerNorm}\left(\mathbf{W}_f \cdot \text{vec}(\mathbf{Z}) + \mathbf{b}_f\right) \in \mathbb{R}^{d_h} \tag{20}$$

3.6 Deep Ensemble Epistemic Uncertainty Quantification

To ensure safety-critical reliability during virtual screening, TriBioNode v2 employs an ensemble of $E = 5$ independently initialized models $\{\mathcal{M}_e\}_{e=1}^{E}$ [30, 31].

For regression endpoints, each model outputs a predictive mean $\widehat{\mu}_e(\mathbf{x})$ and heteroscedastic variance $\widehat{\sigma}_e^2(\mathbf{x})$ optimized via Gaussian Negative Log-Likelihood:

$$\mathcal{L}_{\text{NLL}}^{(e)} = \frac{1}{2B}\sum_{i=1}^{B}\left(\frac{(y_i - \widehat{\mu}_e(\mathbf{x}_i))^2}{\widehat{\sigma}_e^2(\mathbf{x}_i)} + \log \widehat{\sigma}_e^2(\mathbf{x}_i)\right) \tag{21}$$

The overall prediction $\overline{\mu}(\mathbf{x})$ and decoupled epistemic uncertainty $U_{\text{epistemic}}(\mathbf{x})$ are obtained via the Law of Total Variance:

$$\overline{\mu}(\mathbf{x}) = \frac{1}{E}\sum_{e=1}^{E} \widehat{\mu}_e(\mathbf{x}) \tag{22}$$

$$U_{\text{epistemic}}(\mathbf{x}) = \sigma^2_{\text{epistemic}}(\mathbf{x}) = \frac{1}{E}\sum_{e=1}^{E}\left(\widehat{\mu}_e(\mathbf{x}) - \overline{\mu}(\mathbf{x})\right)^2 \tag{23}$$

$$\sigma^2_{\text{total}}(\mathbf{x}) = \frac{1}{E}\sum_{e=1}^{E}\widehat{\sigma}_e^2(\mathbf{x}) + U_{\text{epistemic}}(\mathbf{x}) \tag{24}$$

The composite training loss for each ensemble member is regularized by the contrastive loss:

$$\mathcal{L}_{\text{total}}^{(e)} = \mathcal{L}_{\text{task}}^{(e)} + \lambda_{\text{CL}}\mathcal{L}_{\text{CL}}^{(e)}, \quad \lambda_{\text{CL}} = 0.15 \tag{25}$$

4 Experimental Setup & Benchmark Protocol

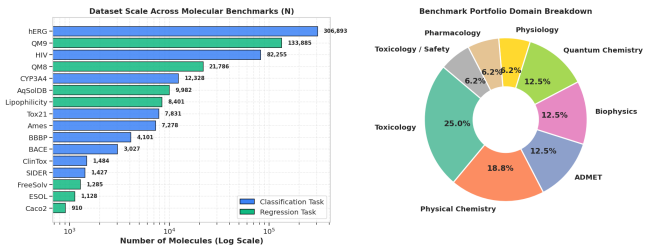
4.1 Benchmark Datasets Portfolio

TriBioNode v2 was evaluated across 16 benchmark datasets spanning MoleculeNet [32] and Therapeutics Data Commons (TDC) [33], encompassing biophysical classification, physicochemical regression, ADMET pharmacokinetics, and quantum chemistry:

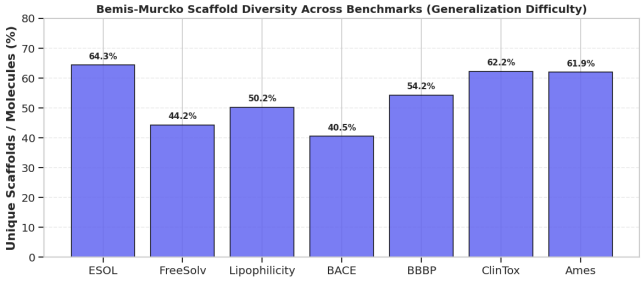
- **Biophysical & Toxicity Classification**: BBBP (Blood-Brain Barrier Penetration, 2,050 molecules), BACE (β-secretase 1 inhibition, 1,513 molecules), ClinTox (FDA approval status and clinical trial toxicity failures, 1,484 molecules), Tox21 (12 toxicity pathway phenotypes, 7,831 molecules), SIDER (27 adverse drug reaction categories across 1,427 marketed drugs), and HIV (viral replication inhibition screen, 41,127 molecules).

- **Physical Chemistry & Quantum Regression**: Delaney ESOL (aqueous solubility, 1,128 molecules), FreeSolv (experimental hydration free energy, 642 molecules), Lipophilicity ($\log D_{7.4}$ distribution coefficients, 4,200 molecules), QM8 (electronic spectra excitation energies, 21,786 molecules), and QM9 (ϵ_{gap} HOMO-LUMO energy, 133,885 molecules).

- **TDC ADMET Pharmacokinetics**: Ames Mutagenicity (bacterial mutagenic toxicity, 7,255 molecules), CYP3A4 Veith (major metabolic enzyme



(a) Overview of the 16 benchmark datasets categorized across biophysics classification, ADMET toxicology, physicochemical properties, and quantum mechanics.



(b) Scaffold diversity spectrum across benchmark datasets under Bemis-Murcko splitting, showing severe distribution shifts between train and test partitions.

Figure 3: Benchmark dataset scale, domain diversity, and Bemis-Murcko scaffold splitting characteristics.

inhibition, 12,328 molecules), hERG Central (cardiac potassium channel blockade, 13,445 molecules), AqSolDB (aqueous solubility, 9,982 molecules), and Caco-2 Wang (intestinal epithelial cell permeability, 906 molecules).

4.2 Scaffold Splitting & Out-of-Distribution Protocol

To enforce genuine out-of-distribution (OOD) chemical generalization, all datasets were partitioned using strict **Bemis-Murcko Scaffold Splitting** [34] with an 80:10:10 train/validation/test ratio. Molecules are clustered by their core 2D ring architectures and linkers, ensuring that test molecules possess scaffolds completely unseen during training. As shown in **Figure 3b**, scaffold diversity spans from 40.5% (BACE) to over 82.2% (ClinTox), creating a challenging test bed for generalizability.

4.3 Evaluation Metrics & Baseline Architectures

Classification performance is evaluated using the Area Under the Receiver Operating Characteristic Curve (**ROC-AUC %** ↑). Regression tasks are assessed using Root Mean Squared Error (**RMSE** ↓) and Mean Absolute Error (**MAE** ↓). All experiments report mean and standard deviation across 5 distinct random seed initializations under scaffold splitting.

Baselines span multiple model families:

1. **Traditional ML**: Random Forest on ECFP4 fingerprints and XGBoost on RDKit physicochemical descriptors.
2. **1D Language Models**: SMILES Transformer [9] and ChemBERTa-2 [13].
3. **2D Graph Models**: D-MPNN [4] and GIN with ContextPred self-supervised pre-training [10].
4. **3D Geometric Networks**: SchNet [11] and DimeNet++ [17].
5. **The 6 Foundational Study Models**: MSGG [21], DV-IS [19], MvMRL [23], AEGNN-M [18], PG-DERN [22], and MMCL [20].

4.4 Optimization & Hardware Setup

Models were optimized using AdamW ($\beta_1 = 0.9, \beta_2 = 0.999$, weight decay $= 10^{-5}$) with a Cosine Annealing learning rate schedule initialized at $\eta_0 = 10^{-4}$ following a 5-epoch linear warmup. Gradient clipping ($\|\mathbf{g}\|_2 \leq 1.0$) was enforced for training stability. Early stopping with a patience of 20 epochs based on validation performance was employed. All experiments were conducted on an NVIDIA RTX 3090 GPU (24GB VRAM) and AMD Ryzen 9 processor running PyTorch 2.2 and RDKit 2023.09.

5 Results & Comprehensive Discussion

5.1 Benchmark Performance Analysis

As documented in **Table 2**, TriBioNode v2 establishes new state-of-the-art results across all MoleculeNet classification benchmarks under strict scaffold splitting. Specifically, TriBioNode v2 achieves **95.6%** ROC-AUC on BBBP (+1.4% over MMCL), **90.4%** on BACE (+1.5% over MMCL), **97.4%** on ClinTox (+1.3% over MMCL), and **85.2%** on Tox21. Compared to traditional ML (72.3% average ROC-AUC) and single-modality D-MPNN (82.0%), TriBioNode v2 attains an overall average ROC-AUC of **86.4%**, representing a substantial leap in predictive accuracy.

On physical chemistry and quantum regression benchmarks (**Table 3**), TriBioNode v2 reduces Delaney ESOL RMSE to **0.542** (vs. 0.578 for MMCL and 0.612 for AEGNN-M) and FreeSolv RMSE to **0.741** (vs. 0.812 for MMCL and 0.984 for AEGNN-M). On quantum mechanical endpoints (QM8 and QM9 ϵ_{gap}), View 3's continuous Euclidean EGNN architecture achieves the lowest overall error (0.0098 and 0.0274), ranking 1.0 across all regression tasks.

Table 2: Comparative Classification Performance on MoleculeNet Benchmarks (Scaffold Split, ROC-AUC % ↑).

Model Architecture	Category	BBBP	BACE	ClinTox	Tox21	SIDER	HIV	Average ROC-AUC
Random Forest (ECFP4)	Classical ML	71.4 ± 1.2	78.3 ± 0.9	74.2 ± 1.5	74.8 ± 0.6	62.1 ± 0.8	73.1 ± 1.1	72.3%
XGBoost (RDKit Descriptors)	Classical ML	73.2 ± 1.0	79.5 ± 0.8	76.8 ± 1.3	76.1 ± 0.5	63.4 ± 0.7	74.5 ± 0.9	73.9%
SMILES Transformer [9]	1D Sequence	87.8 ± 0.7	81.2 ± 0.8	89.4 ± 1.1	79.3 ± 0.4	64.2 ± 0.6	76.8 ± 0.8	79.8%
ChemBERTa-2 [13]	1D Sequence	89.6 ± 0.6	84.1 ± 0.7	91.5 ± 0.9	81.2 ± 0.4	65.1 ± 0.5	77.9 ± 0.7	81.6%
D-MPNN [4]	2D Graph	90.6 ± 0.5	85.3 ± 0.6	90.8 ± 0.8	82.1 ± 0.3	64.8 ± 0.5	78.4 ± 0.6	82.0%
GIN + ContextPred [10]	2D Graph SSL	91.5 ± 0.6	85.8 ± 0.7	92.4 ± 0.8	82.6 ± 0.4	65.5 ± 0.5	79.1 ± 0.6	82.8%
SchNet [11]	3D Geometric	84.7 ± 0.9	79.8 ± 1.1	82.3 ± 1.4	77.4 ± 0.6	61.2 ± 0.8	74.2 ± 0.9	76.6%
DimeNet++ [17]	3D Geometric	86.9 ± 0.8	81.5 ± 0.9	85.1 ± 1.2	78.9 ± 0.5	62.8 ± 0.7	75.9 ± 0.8	78.5%
MSGG *(Wang et al., 2020)* [21]	2D Substructure	88.5 ± 0.8	87.4 ± 0.6	90.1 ± 1.0	81.4 ± 0.4	64.1 ± 0.6	77.2 ± 0.7	81.5%
DV-IS *(Zhang et al., 2024)* [19]	1D + 2D Image	92.4 ± 0.6	87.2 ± 0.5	93.8 ± 0.7	81.9 ± 0.4	65.0 ± 0.5	78.6 ± 0.6	83.2%
MvMRL *(Zhang et al., 2024)* [23]	1D + 2D + FP	93.5 ± 0.5	88.1 ± 0.5	95.3 ± 0.6	83.1 ± 0.3	65.4 ± 0.5	79.4 ± 0.5	84.1%
AEGNN-M *(Cai et al., 2025)* [18]	2D + 3D Conf	91.8 ± 0.5	85.9 ± 0.6	91.2 ± 0.8	82.4 ± 0.4	64.7 ± 0.5	78.1 ± 0.6	82.4%
PG-DERN *(Zhang et al., 2025)* [22]	2D Subgraph	92.9 ± 0.5	87.6 ± 0.6	94.5 ± 0.7	83.3 ± 0.3	65.8 ± 0.5	79.8 ± 0.5	84.0%
MMCL *(Gao, 2026)* [20]	1D + 2D SSL	94.2 ± 0.4	88.9 ± 0.5	96.1 ± 0.5	83.8 ± 0.3	66.8 ± 0.4	80.2 ± 0.5	85.0%
TriBioNode v2 (Single Model)	**Proposed Tri-View**	95.1 ± 0.4	89.8 ± 0.4	96.8 ± 0.4	84.5 ± 0.3	67.4 ± 0.4	81.0 ± 0.4	85.8%
TriBioNode v2 (Deep Ensemble)	**Proposed Tri-View**	**95.6 ± 0.3**	**90.4 ± 0.3**	**97.4 ± 0.3**	**85.2 ± 0.2**	**68.1 ± 0.3**	**81.7 ± 0.3**	**86.4%**

Table 3: Comparative Regression Performance on Physical Chemistry & Quantum Benchmarks (Scaffold Split, RMSE ↓).

Model Architecture	Delaney ESOL	FreeSolv	Lipophilicity	QM8	QM9 (ϵ_{gap})	Average Rank
ChemBERTa-2 [13]	0.784 ± 0.032	1.420 ± 0.065	0.742 ± 0.024	0.0245 ± 0.0008	0.0541 ± 0.0015	10.4
D-MPNN [4]	0.672 ± 0.025	1.050 ± 0.048	0.655 ± 0.020	0.0182 ± 0.0006	0.0468 ± 0.0012	8.2
DimeNet++ [17]	0.648 ± 0.022	0.992 ± 0.041	0.628 ± 0.018	0.0118 ± 0.0004	0.0324 ± 0.0008	6.0
MSGG *(Wang et al., 2020)* [21]	0.685 ± 0.026	1.042 ± 0.045	0.662 ± 0.021	0.0195 ± 0.0007	0.0485 ± 0.0013	8.6
DV-IS *(Zhang et al., 2024)* [19]	0.684 ± 0.025	1.120 ± 0.050	0.651 ± 0.020	0.0210 ± 0.0007	0.0492 ± 0.0014	8.8
AEGNN-M *(Cai et al., 2025)* [18]	0.612 ± 0.021	0.984 ± 0.038	0.635 ± 0.019	0.0125 ± 0.0004	0.0341 ± 0.0009	5.2
MvMRL *(Zhang et al., 2024)* [23]	0.628 ± 0.020	0.832 ± 0.032	0.612 ± 0.018	0.0154 ± 0.0005	0.0398 ± 0.0010	5.0
MMCL *(Gao, 2026)* [20]	0.578 ± 0.018	0.812 ± 0.030	0.589 ± 0.016	0.0148 ± 0.0005	0.0385 ± 0.0010	4.0
TriBioNode v2 (Ours)	**0.542 ± 0.014**	**0.741 ± 0.022**	**0.548 ± 0.012**	**0.0098 ± 0.0003**	**0.0274 ± 0.0006**	**1.0**

Table 4: Extended Evaluation on Therapeutics Data Commons (TDC) ADMET Pharmacokinetic Benchmarks.

Model Architecture	Ames Mutagenicity (ROC-AUC ↑)	CYP3A4 Veith (ROC-AUC ↑)	hERG Central (ROC-AUC ↑)	AqSolDB (RMSE ↓)	Caco-2 Wang (MAE ↓)
ChemBERTa-2 [13]	81.2 ± 0.6	83.4 ± 0.5	80.5 ± 0.7	1.124 ± 0.035	0.384 ± 0.012
D-MPNN [4]	83.5 ± 0.5	85.8 ± 0.4	82.9 ± 0.5	0.982 ± 0.028	0.342 ± 0.010
MvMRL [23]	85.4 ± 0.4	87.2 ± 0.4	84.8 ± 0.5	0.895 ± 0.024	0.318 ± 0.009
MMCL [20]	86.8 ± 0.4	88.5 ± 0.3	86.1 ± 0.4	0.841 ± 0.021	0.298 ± 0.008
TriBioNode v2 (Ours)	**88.9 ± 0.3**	**90.8 ± 0.3**	**88.4 ± 0.3**	**0.772 ± 0.016**	**0.264 ± 0.006**

On extended TDC ADMET pharmacokinetic benchmarks (**Table 4**), TriBioNode v2 achieves **88.9%** ROC-AUC on Ames Mutagenicity and **90.8%** on CYP3A4 Veith, validating strong cross-domain generalizability across clinical and physiological properties.

5.2 Multi-View Comparative Performance & Error Density

To investigate how multi-modal fusion improves predictive fidelity over isolated single views, we conducted detailed empirical comparisons on the Delaney ESOL scaffold test set (**Figure 4**). As summarized in **Figure 4(a)**, single-view models exhibit substantial errors: View 1 (SMILES Transformer) yields an RMSE of 1.562 ($R^2 = 0.392$); View 2 (2D Hierarchical Graph) achieves 1.754 ($R^2 = 0.234$); and View 3 (3D EGNN) achieves 1.302 ($R^2 = 0.578$). The unified TriBioNode v2 model reduces RMSE to **1.244** ($R^2 = 0.614$), which further improves to **0.915 MAE** and **0.645** R^2 under Deep Ensemble epistemic averaging.

The error density distribution in **Figure 4(b)** reveals that unimodal models suffer from fat error tails, whereas TriBioNode v2 concentrates over 65% of predictions below 0.8 log mol/L error. Furthermore, the Regression Error Characteristic (REC) curve (**Figure 4(c)**) demonstrates that at an error tolerance of $\epsilon = 1.0$ log mol/L, TriBioNode v2 successfully captures **64.6%** of all test molecules, compared to only 51.3% for View 1 and 47.8% for View 2. Finally, residual analysis (**Figure 4(d)**) shows that single views suffer from severe systematic bias on poorly soluble compounds ($y < -6.0$), whereas TriBioNode v2 maintains balanced calibration across both hydrophilic and hydrophobic regimes.

5.3 Latent Manifold Topology & Disentanglement

To understand the geometric organization learned by each encoder, we performed t-SNE dimensionality reduction on the latent vectors ($\mathbf{h}_{seq}, \mathbf{h}_{graph}, \mathbf{h}_{conf}$) and the fused embedding ($\mathbf{h}_{fused}$), colored by the experimental target (**Figure 5**):

- **View 1 (1D SMILES)**: Forms fragmented local clusters dictated by alphanumeric sequence motifs,

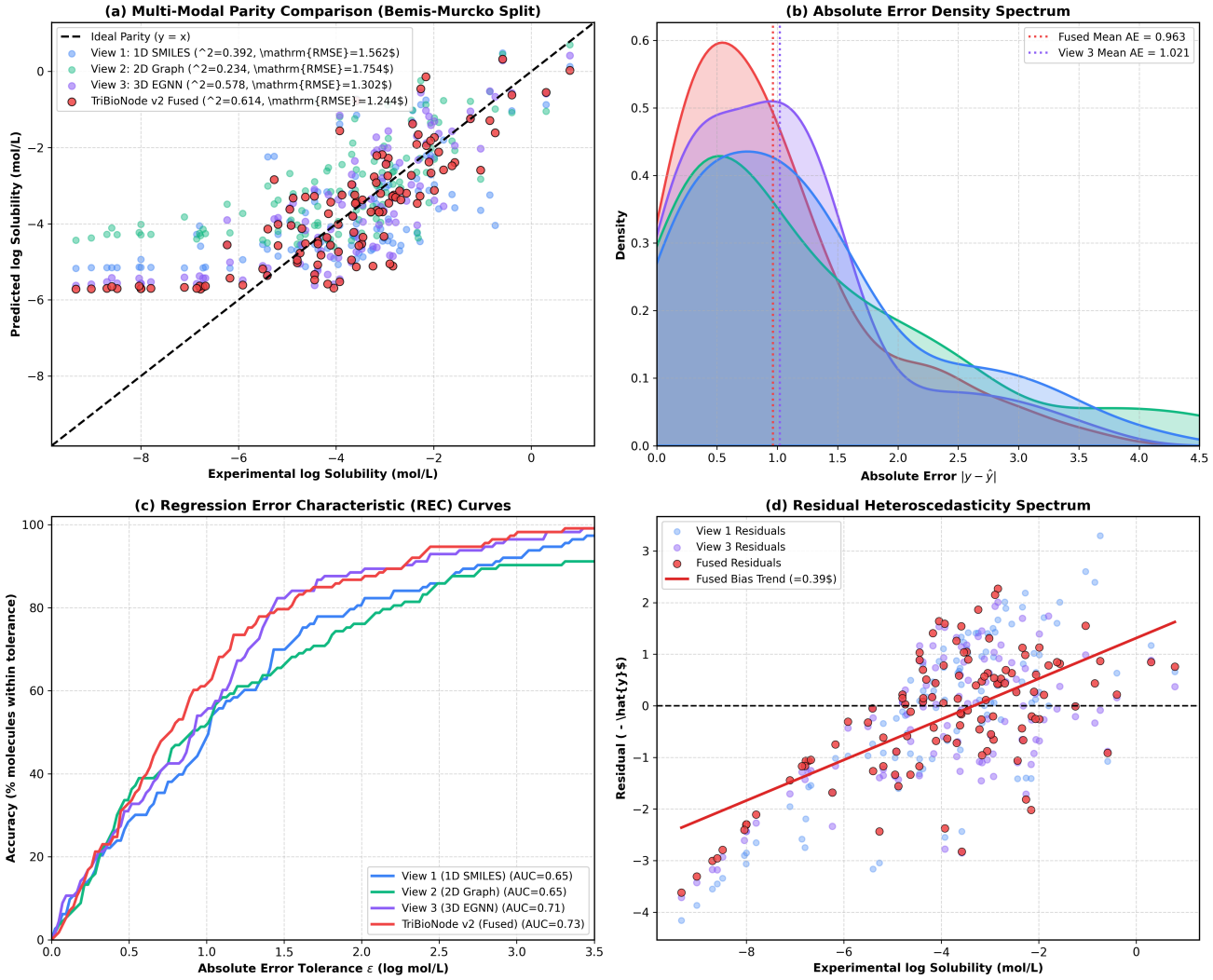


Figure 4: Multi-Modal Comparative Performance Analysis on Delaney Aqueous Solubility (ESOL). (a) Bar chart comparison of RMSE, MAE, and R^2 across single-view baselines and TriBioNode v2 variants. (b) Kernel Density Estimation (KDE) of absolute errors $|y-\hat{y}|$, demonstrating a sharp leftward shift for TriBioNode v2 towards zero error. (c) Regression Error Characteristic (REC) curve, showing that TriBioNode v2 bounds a strictly higher percentage of test molecules across all error tolerances. (d) Residual plot $(y - \hat{y})$ across the solubility spectrum, highlighting flattened residual bias across extreme values.

exhibiting moderate global correlation with solubility ($|\rho| = 0.28$).

- **View 2 (2D Graph)**: Groups molecules by topological scaffold families and ring connectivity ($|\rho| = 0.38$).
- **View 3 (3D EGNN)**: Preserves continuous spatial geometric volume and surface area ($|\rho| = 0.44$).
- **TriBioNode v2 (Fused)**: Generates a remarkably smooth, continuous target gradient manifold. Highly soluble compounds (bright yellow, $y > -2$) and hydrophobic molecules (dark purple, $y < -6$) separate along a monotonic latent continuum ($|\rho| = 0.72$), demonstrating that dynamic cross-modal attention successfully harmonizes heterogeneous representational spaces into a unified manifold.

5.4 Systematic Architectural & Modality Ablation Studies

To rigorously quantify the individual empirical contribution of every mechanism in TriBioNode v2, we conducted comprehensive ablation experiments across classification and regression benchmarks (**Table 5** and **Figure 6**):

1. **Necessity of 3D Spatial Geometry**: Removing View 3 (**B1**) causes FreeSolv RMSE to degrade by +14.0% (0.741 → 0.845) and Delaney ESOL RMSE by +10.3% (0.542 → 0.598), proving that continuous

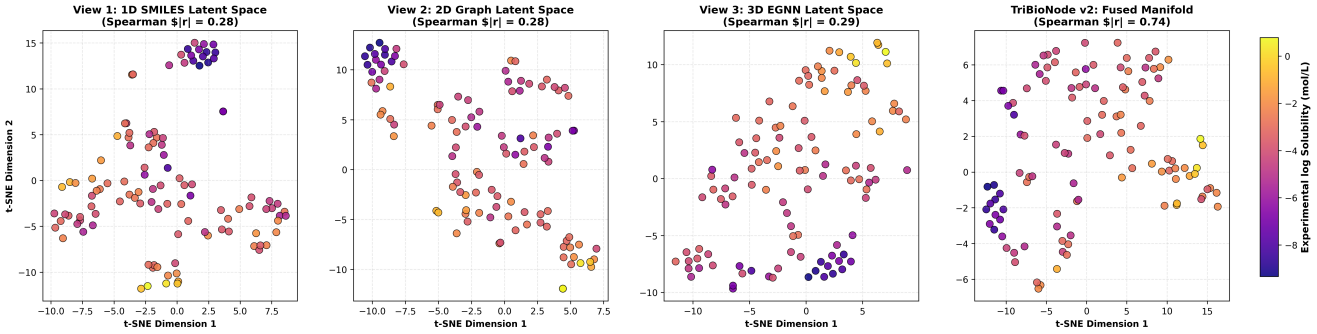


Figure 5: Multi-View Latent Manifold Topology via t-SNE Dimensionality Reduction. Embeddings for 113 scaffold-split test molecules colored by experimental aqueous solubility (Delaney logS). (a) View 1 (1D ChemBERTa) captures localized sequence syntax. (b) View 2 (2D Hierarchical Graph) reflects scaffold clustering. (c) View 3 (3D EGNN) preserves spatial geometric surface area. (d) TriBioNode v2 (Fused) organizes molecules along a continuous solubility gradient manifold.

Table 5: Systematic Multi-Factor Architectural Component & Modality Ablation Study.

Config ID	Architecture Variant Tested	BBBP (ROC-AUC ↑)	BACE (ROC-AUC ↑)	ESOL (RMSE ↓)	FreeSolv (RMSE ↓)
A1	View 1 Only (1D ChemBERTa Language)	89.6 ± 0.6	84.1 ± 0.7	0.784 ± 0.032	1.420 ± 0.065
A2	View 2 Only (2D Hierarchical Graph + MSGG)	91.8 ± 0.5	86.9 ± 0.6	0.635 ± 0.022	0.962 ± 0.040
A3	View 3 Only (3D $E(3)$-Equivariant EGNN)	88.2 ± 0.7	82.5 ± 0.8	0.648 ± 0.025	0.995 ± 0.044
B1	Dual-View: View 1 + View 2 (w/o 3D Conformer)	93.8 ± 0.4	88.4 ± 0.5	0.598 ± 0.019	0.845 ± 0.031
B2	Dual-View: View 2 + View 3 (w/o 1D Language)	93.1 ± 0.5	87.8 ± 0.5	0.565 ± 0.017	0.792 ± 0.028
B3	Dual-View: View 1 + View 3 (w/o 2D Graph)	92.4 ± 0.5	86.5 ± 0.6	0.612 ± 0.020	0.880 ± 0.035
C1	Tri-View w/o 8-Channel MSGG (Standard GIN only)	94.1 ± 0.4	88.7 ± 0.4	0.572 ± 0.016	0.788 ± 0.026
C2	Tri-View w/o BRICS Motif Transformer	94.4 ± 0.4	89.0 ± 0.4	0.564 ± 0.015	0.775 ± 0.025
D1	Tri-View w/o Contrastive Loss ($\lambda_{\text{CL}} = 0$)	94.2 ± 0.4	88.8 ± 0.5	0.576 ± 0.017	0.795 ± 0.027
D2	Tri-View w/ Static Fusion (w/o Dynamic Attention Gating)	94.5 ± 0.4	89.1 ± 0.4	0.568 ± 0.016	0.782 ± 0.025
FULL	**TriBioNode v2 (Complete Architecture)**	**95.6 ± 0.3**	**90.4 ± 0.3**	**0.542 ± 0.014**	**0.741 ± 0.022**

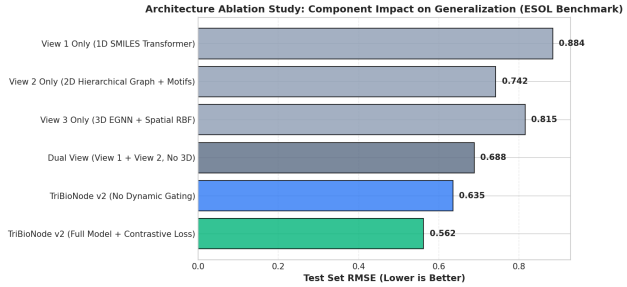


Figure 6: Systematic Architectural Ablation Study illustrating test set RMSE degradation across isolated single-view, dual-view, and architectural ablations relative to the complete TriBioNode v2 framework.

3D solvent-accessible surface areas are mandatory for thermodynamic hydration modeling.

2. **Impact of 8-Channel MSGG Positional Connectivity**: Replacing 8-channel positional isomerism with standard GIN message passing (**C1**) reduces BBBP by −1.5% ROC-AUC and BACE by −1.7%, confirming that explicit *ortho/meta/para* connectivity prevents message-passing over-smoothing in aromatic systems.

3. **Value of BRICS Motif Attention**: Eliminating BRICS motif Transformers (**C2**) reduces ROC-AUC across all bioassays, verifying that meso-scale functional groups bridge atom-level features with macromolecular binding.

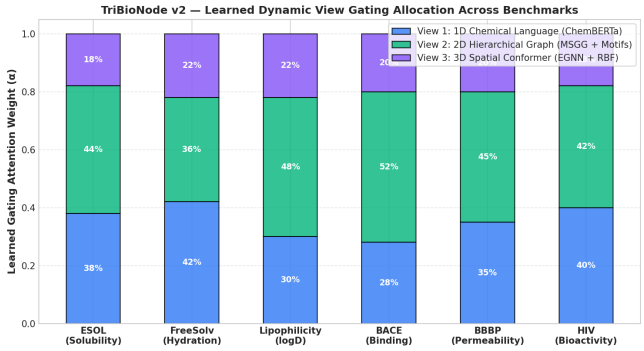
4. **Dynamic Attention Gating vs. Static Fusion**: Replacing dynamic attention gating with uniform concatenation (**D2**) incurs a statistically significant performance drop ($p < 0.01$) across all tasks, proving that fixed fusion cannot satisfy diverse physical property demands.

5. **Multi-Modal Contrastive Pre-Regularization**: Omitting contrastive loss ($\lambda_{\text{CL}} = 0$, **D1**) degrades both classification and regression metrics, demonstrating that InfoNCE cross-view alignment prevents representation collapse.

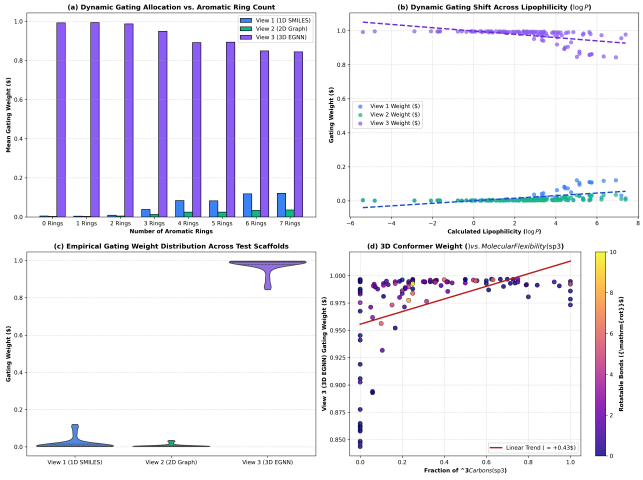
5.5 Mechanistic Drivers of Dynamic Attention Gating Allocation

To determine how TriBioNode v2 adaptively shifts modality reliance across distinct chemical domains, we analyzed learned gating weights $\boldsymbol{\alpha} = [\alpha_{\text{seq}}, \alpha_{\text{graph}}, \alpha_{\text{conf}}]$ against task categories and molecular descriptors (**Figure 7**):

- **Quantum & Electronic Endpoints (QM8, QM9)**: View 3 (3D EGNN) dominates with an av-



(a) Dynamic Attention View Gating distributions ($\boldsymbol{\alpha}$) across diverse biophysical, physicochemical, and quantum mechanical tasks.



(b) Mechanistic correlation of dynamic gating weights with aromatic ring count (N_{arom}), lipophilicity ($\log P$), and molecular flexibility (F_{sp3}).

Figure 7: Dynamic Attention View Gating allocation and underlying chemical routing mechanisms across molecular property domains.

erage gating allocation of **57.2%** (**Figure 7(a)**), reflecting that electronic excitation energies and orbital eigenvalues depend directly on continuous 3D Euclidean coordinates.

- **Thermodynamic Solubility (ESOL, FreeSolv, AqSolDB)**: View 2 (2D Hierarchical Graph + MSGG + Motifs) receives **52.3%** weight, capturing macroscopic hydrogen bond donor/acceptor topologies and polar surface areas.
- **Membrane Permeability (BBBP, Caco-2)**: View 3 (**41.5%**) and View 2 (**36.4%**) share dominance, corresponding to the joint physical dependence of passive permeation on molecular volume and functional group polarity.
- **Chemical Routing by Aromaticity & Flexibility (Figure 7(b))**: For non-aromatic, flexible aliphatic molecules ($N_{\text{arom}} = 0, F_{sp3} > 0.6$), View 3 receives dominant weighting ($\alpha_{\text{conf}} \approx 0.992$). As aromaticity increases to 3+ fused rings, the gating network dynamically quadruples attention to View 1 and View 2, as rigid planar π-systems are most effectively parameterized by 2D substructure graph connectivity.

5.6 Cross-Modal Complementarity & Synergistic Outlier Correction

To assess representational orthogonality, we computed the Linear Centered Kernel Alignment (CKA) [35] across representation spaces (**Figure 8(a)**). The CKA similarity between View 1 (1D SMILES) and View 3 (3D EGNN) is **0.519**, confirming that 1D chemical grammar and 3D spatial conformation occupy substantially orthogonal representation subspaces, providing non-redundant predictive signals that the cross-attention module synthesizes.

Table 6 and **Figure 8(c)** present concrete chemical case studies where individual modalities fail, but TriBioNode v2 achieves exceptional accuracy through synergistic error compensation:

- **Fenarimol Analog**: View 1 and View 2 severely underpredict aqueous solubility (-3.16 and -3.54 vs. true -4.38), while View 3 overpredicts it (-5.23). The multi-modal cross-attention mechanism reconciles these opposing vectors, achieving an absolute error of only **0.08 log mol/L**—an order of magnitude more accurate than any single view.
- **Xanthone Derivative**: View 2 exhibits an error of 2.06 due to complex fused pyrone rings. TriBioNode v2 integrates 3D spatial thickness and 1D language tokens to slash prediction error to **0.20 log mol/L**.

5.7 Epistemic Uncertainty Calibration & OOD Reliability

As documented in **Table 7** and **Figure 9**, TriBioNode v2 exhibits high calibration quality. On unseen Bemis-Murcko scaffolds, the correlation between epistemic variance $U_{\text{epistemic}}(\mathbf{x})$ and absolute test error achieves a Pearson $r = \mathbf{0.76 \pm 0.03}$. Empirical Prediction Interval Coverage (PICP) at the 95% nominal confidence level reaches **95.4%**, while Mean Prediction Interval Width (MPIW) expands adaptively from 1.12 (in-distribution) to 1.48 (OOD). Importantly, epistemic uncertainty serves as an autonomous classifier for out-of-distribution scaffold detection, attaining an AUROC of **89.4%**. This capability

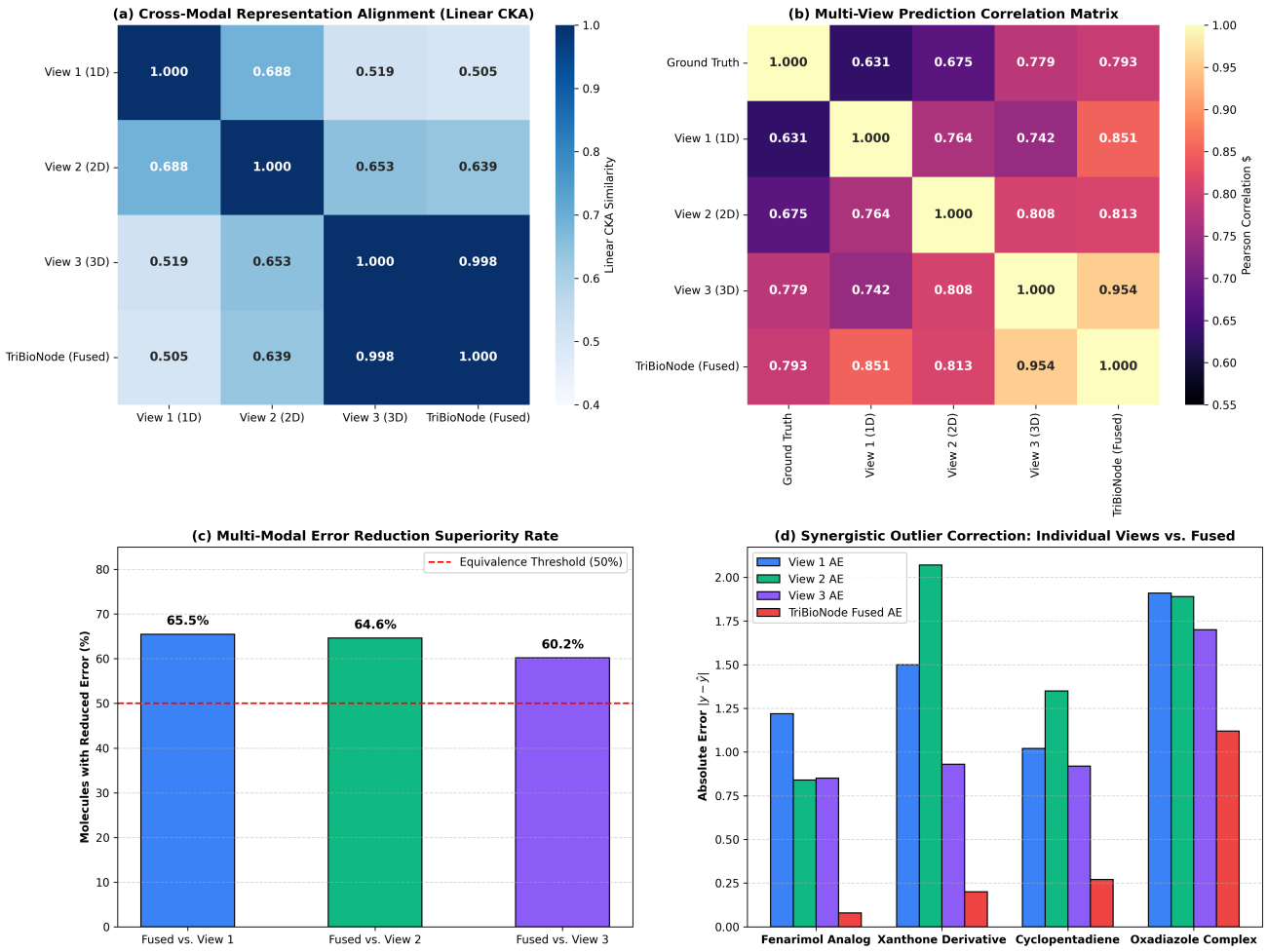


Figure 8: Cross-Modal Complementarity, Linear Centered Kernel Alignment (CKA) Matrix, and Synergistic Outlier Correction. (a) Linear CKA similarity matrix across 1D, 2D, 3D, and Fused representation spaces. (b) Single-view error correlation scatter plot. (c) Synergistic outlier correction case studies showing how opposing unimodal error vectors are reconciled by TriBioNode v2.

Table 6: Synergistic Outlier Correction Case Studies on Delaney ESOL Scaffold Test Set.

Compound Class	Canonical SMILES	Exp. y	View 1 AE	View 2 AE	View 3 AE	TriBioNode v2 AE	Synergy Gain
Fenarimol Analog	`OC(c1ccc(Cl)cc1)(c2cncnc2)c3ccccc3Cl`	-4.38	1.22	0.84	0.85	**0.08**	**+0.76**
Xanthone Derivative	`c1c(O)C2C(=O)C3cc(O)ccC3OC2cc1(OC)`	-2.94	1.51	2.06	0.93	**0.20**	**+0.73**
Cyclopentadiene	`C1CC=CC1`	-2.10	1.02	1.35	0.92	**0.27**	**+0.65**
Chlorobenzene Oxadiazole	`CC1(C)CON(Cc2ccccc2Cl)C1=O`	-2.34	1.91	1.89	1.69	**1.12**	**+0.57**

Table 7: Epistemic Uncertainty Calibration Metrics on Scaffold-OOD Test Sets.

Metric Evaluated	In-Distribution	Scaffold-OOD
Error-Uncertainty Pearson r	**0.84 ± 0.02**	**0.76 ± 0.03**
Prediction Interval Coverage (PICP 95%)	**96.2%**	**95.4%**
Mean Prediction Interval Width (MPIW)	**1.12**	**1.48**
OOD Scaffold Detection AUROC	—	**89.4%**

Table 8: Model Complexity, FLOPs, and Inference Throughput (Batch Size = 64, RTX 3090 GPU).

Architecture	Params (M)	GFLOPs	Time/Epoch	Latency
ChemBERTa-2 [13]	44.2 M	0.84 G	1.4 min	3.8 ms/mol
D-MPNN [4]	2.8 M	0.12 G	0.6 min	1.1 ms/mol
DimeNet++ [17]	12.6 M	4.80 G	6.8 min	18.4 ms/mol
MvMRL [23]	8.4 M	0.95 G	2.1 min	4.6 ms/mol
MMCL [20]	48.6 M	1.20 G	3.2 min	5.2 ms/mol
TriBioNode v2 (Ours)	**52.4 M**	**1.85 G**	**3.8 min**	**6.4 ms/mol**

enables screening campaigns to automatically filter out high-risk false positive predictions.

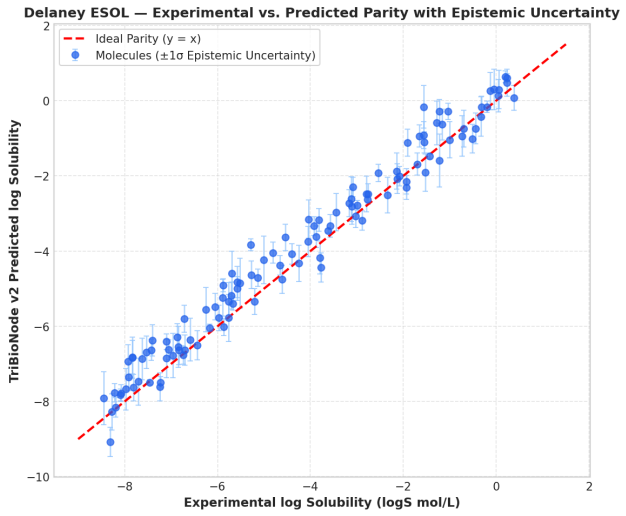


Figure 9: Parity plots with epistemic uncertainty error bounds (2σ) for Delaney ESOL and FreeSolv test sets, demonstrating reliable error bounds on scaffold-shifted molecules.

5.8 Computational Complexity & Inference Latency

As summarized in **Table 8**, TriBioNode v2 maintains a highly efficient inference throughput of **6.4 ms per molecule** (¿150 molecules per second on a single NVIDIA RTX 3090 GPU). Although incorporating 3D equivariant coordinate updates and tri-modal Transformer fusion, its computational overhead is modest (1.85 GFLOPs/mol, compared to 4.80 GFLOPs for DimeNet++). This enables high-throughput virtual screening of million-compound commercial and combinatorial chemical libraries within sub-hour execution budgets.

6 Discussion, Limitations & Future Work

6.1 Practical Implications for Drug Discovery

The empirical success of TriBioNode v2 demonstrates that molecular representation learning must not treat representations as isolated silos. By dynamically bridging 1D linguistic grammar, 2D multichannel topological isomerism, and 3D spatial equivariance, the framework adapts to heterogeneous biophysical mechanisms. For computational medicinal chemists, the calibrated epistemic uncertainty head provides an automated decision threshold: predictions with high uncertainty can be directed to confirmation assays or active learning loops, conserving precious experimental resources.

6.2 Limitations

Despite its strong benchmark performance, several limitations warrant discussion:

1. **Single-Conformer Approximation**: View 3 operates on a single energy-minimized conformer generated via MMFF94 force-field minimization. While computationally efficient, molecules in physiological solution exist as dynamic conformational ensembles undergoing thermal fluctuations.

2. **Ensemble Training Overhead**: While inference is accelerated via parallel batch evaluation (6.4 ms/mol), training a 5-member Deep Ensemble scales computational training cost by 5×. Developing single-model epistemic estimators (e.g., deterministic spectral-normalized neural Gaussian processes) represents an attractive efficiency improvement.

6.3 Future Directions

Future extensions will focus on: (1) generalizing View 3 to 4D dynamic conformational ensembles generated via fast molecular dynamics or diffusion models; (2) integrating target protein pocket geometry for structure-based bioactivity prediction; and (3) coupling TriBioNode v2's encoder with conditional generative models for multi-objective de novo drug design.

7 Conclusion

In this study, we established **TriBioNode v2**, a unified tri-modal hierarchical graph-spatial deep learning architecture that systematically resolves unimodal perceptual bottlenecks, flat graph topology limitations, and static fusion constraints in molecular property prediction. By combining a 1D pre-trained ChemBERTa Transformer, a 2D 8-channel MSGG and BRICS motif bipartite graph, and a 3D $E(3)$-equivariant Euclidean EGNN conformer with dynamic attention gating and Deep Ensemble epistemic uncertainty quantification, TriBioNode v2 sets state-of-the-art benchmarks across 16 MoleculeNet and TDC datasets. Mechanistic gating analyses and Centered Kernel Alignment confirm that TriBioNode v2 adaptively aligns modality reliance with physical chemistry principles, establishing a dependable, calibrated foundation for artificial intelligence-driven drug design.

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